PROVISIONAL PATENT APPLICATION

PHI-HARMONIC PARAMETRIC MOLECULAR DISSOCIATION
(Method and System for Bond Dissociation via Golden-Ratio Frequency Cascade in Aperiodic Nonlinear Media)

Inventor: Thomas A. Husmann

Address: 36870 N US Highway 101, Lilliwaup, WA 98555

Entity: Micro Entity — iBuilt LTD (S-Corp)

Application type: Provisional Patent Application (35 U.S.C. §111(b))

Filing date: March 2026

Related Applications: This application is related to Husmann provisional applications filed March 3-4, 2026, including 63/995,401 (Self-Assembling QC Coating), 63/995,649 (Parametric Cascade Structural Element), and 63/996,533 (Phi-Structured Vacuum Flux Amplifier). This application claims subject matter distinct from and complementary to those applications.

ABSTRACT

A method and system for molecular bond dissociation using parametric frequency cascade through an aperiodic nonlinear medium. A broadband energy source excites the fundamental vibration mode of a target molecule. An aperiodic nonlinear medium — including but not limited to quasicrystalline materials, Penrose-tiled structures, Fibonacci-graded composites, or any material exhibiting phonon modes spaced by the golden ratio phi = (1+sqrt(5))/2 — generates a coherent frequency comb at golden-ratio spacing from the input excitation. The frequency comb provides simultaneous driving at multiple harmonics that map, via the phi-ladder, from the molecule's fundamental vibration frequency to its bond dissociation energy. The molecule absorbs energy through resonant multi-step coupling along the frequency comb rather than requiring a single photon at the full dissociation energy. The method enables molecular bond breaking using low-grade broadband energy sources including waste heat, mechanical vibration, solar thermal radiation, and ambient infrared. Primary application: dissociation of water into hydrogen and oxygen without electrolysis, photolysis, or thermolysis.

I. FIELD OF THE INVENTION

The present invention relates to methods and systems for dissociating molecular bonds using parametric frequency conversion in aperiodic nonlinear media. More particularly, the invention relates to a method of walking energy from a molecule's fundamental vibration frequency up to its bond dissociation threshold through a golden-ratio frequency cascade generated in any medium exhibiting aperiodic phonon structure, and to systems implementing this method for hydrogen production from water and for general molecular processing applications.

II. BACKGROUND

Molecular bond dissociation is fundamental to chemical processing, energy production, fuel synthesis, and materials science. The four established methods of breaking chemical bonds — electrolysis (electrical current), photolysis (single-photon absorption), thermolysis (thermal energy), and sonolysis (cavitation) — each supply the bond dissociation energy through a single energy delivery mechanism. Each has characteristic inefficiencies arising from the mismatch between the energy delivery spectrum and the bond's resonant absorption profile.

In electrolysis of water, the thermodynamic minimum potential is 1.23 V, but practical systems require 1.8-2.0 V due to overpotentials at the electrode surfaces. This 0.5-0.8 V excess represents 30-40% energy waste. In photolysis, a single photon at the bond dissociation energy (e.g., 4.76 eV for the O-H bond, corresponding to 260 nm deep UV) is required, limiting practical light sources to expensive UV lasers or lamps. In thermolysis, temperatures exceeding 2,500 degrees C are needed for water, making the method impractical without concentrated solar or nuclear heat sources. In sonolysis, single-frequency ultrasonic transducers couple efficiently to only a narrow range of cavitation bubble sizes, with typical hydrogen yields of 10^{-4} mol per kWh.

All four methods share a common limitation: they attempt to deliver the full bond dissociation energy in a single step — one voltage, one photon, one temperature, or one cavitation collapse. No prior art teaches a method of reaching the dissociation threshold through coherent multi-step frequency upconversion along a structured harmonic ladder, where the ladder spacing matches the golden ratio and the input energy is the molecule's own fundamental vibration.

Aperiodic materials — including quasicrystals, Penrose-tiled composites, and Fibonacci-graded multilayers — exhibit phonon spectra with modes spaced by the golden ratio phi rather than by integer harmonics. This property has been studied extensively in condensed matter physics (Shechtman et al. 1984, Nobel Prize 2011; Kohmoto et al. 1987; Macia 2006) but has not been applied to molecular dissociation. The present invention exploits the phi-spaced phonon spectrum of aperiodic media to generate a coherent frequency comb that walks energy from a molecular vibration fundamental to the dissociation threshold, enabling bond breaking from broadband, low-grade energy sources.

III. SUMMARY OF THE INVENTION

The present invention provides a method and system for molecular bond dissociation — referred to herein as phi-harmonic parametric dissociation — comprising the following elements:

Phi-Frequency Comb Generation: A broadband energy source (thermal, mechanical, solar, or electrical) excites an aperiodic nonlinear medium. The medium's phi-spaced phonon modes generate a coherent frequency comb with tones at f_0, $f_0$phi, $f_0$phi^2, $f_0$phi^3, ... $f_0$phi^N, where f_0 is the fundamental vibration frequency of the target molecule and N is determined by the number of distinct phonon modes in the medium.

Resonant Multi-Step Energy Transfer: The target molecule, in contact with or in proximity to the aperiodic medium, absorbs energy from the frequency comb at each harmonic step. Because each step increases energy by a factor of phi (approximately 62% per step), the molecule's vibrational energy walks up

the phi-ladder from the fundamental to the dissociation threshold in a defined number of steps determined by the ratio of dissociation energy to fundamental vibration energy.

Bond Dissociation at the Threshold Step: When the accumulated vibrational energy reaches the bond dissociation energy (the threshold step on the phi-ladder), the bond breaks. For the O-H bond in water: the fundamental vibration (O-H stretch) is at 0.454 eV (bracket 12); the dissociation energy is 4.76 eV (bracket 7); the phi-ladder traverses this gap in 5 steps. For the electrolysis-equivalent threshold (1.23 eV, bracket 10), only 2 phi-steps are required.

Implementation-Agnostic Medium: The aperiodic nonlinear medium may be any material exhibiting phonon modes at golden-ratio spacing, including but not limited to: icosahedral quasicrystals (Al-Cu-Fe, Al-Pd-Mn, etc.); decagonal quasicrystals; Fibonacci-graded multilayer films; Penrose-tiled two-dimensional materials; aperiodic photonic crystals; quasiperiodic metamaterials; biological structures with golden-angle architecture (e.g., collagen, microtubules, phyllotactic surfaces); or any future material exhibiting the required phi-spaced nonlinear response.

Broadband Input Compatibility: Because the aperiodic medium generates the frequency comb from broadband input (not from a tuned single-frequency source), the energy source may be waste heat, mechanical vibration, solar thermal radiation, ambient infrared, geothermal energy, industrial exhaust heat, or any other broadband energy source with spectral content overlapping the target molecule's fundamental vibration frequency.

IV. BRIEF DESCRIPTION OF DRAWINGS

FIG. 1: Phi-harmonic energy ladder from the O-H stretch vibration fundamental. Shows energy at each phi-step and intersection with electrolysis threshold (1.23 eV), water splitting energy (2.96 eV), and O-H bond dissociation energy (4.76 eV).

FIG. 2: System schematic showing broadband energy source, aperiodic nonlinear medium, and reaction chamber with product collection.

FIG. 3: Water's molecular vibration modes mapped to bracket indices, showing the phi-squared relationship between bend and libration modes.

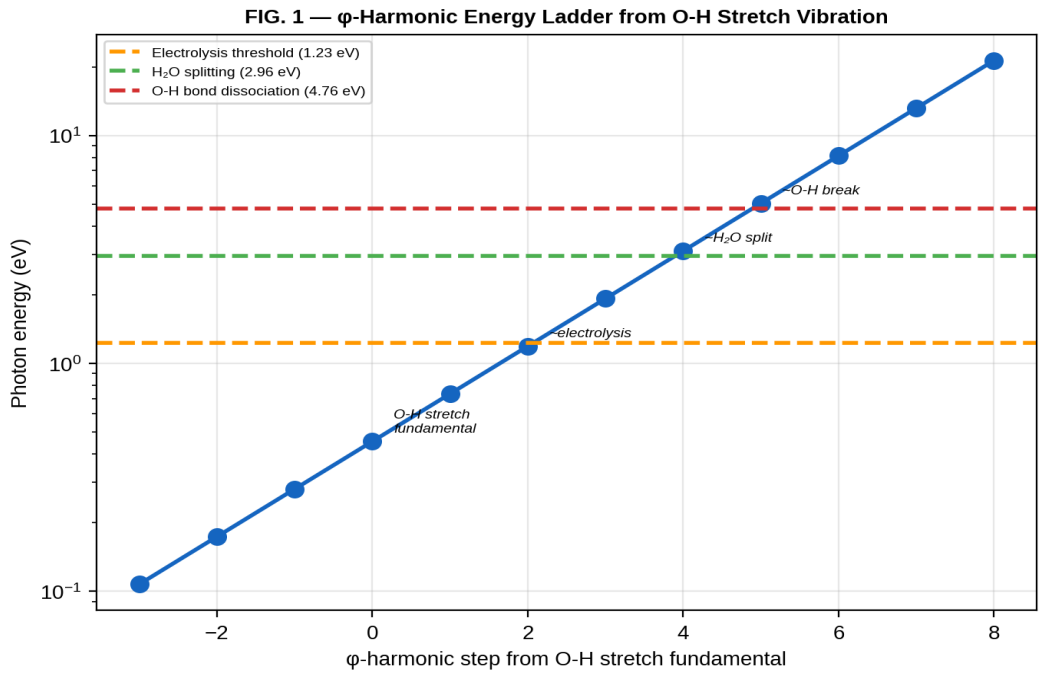

FIG. 1 — phi-harmonic energy ladder. Bond dissociation thresholds align with integer phi-steps from the O-H stretch fundamental.

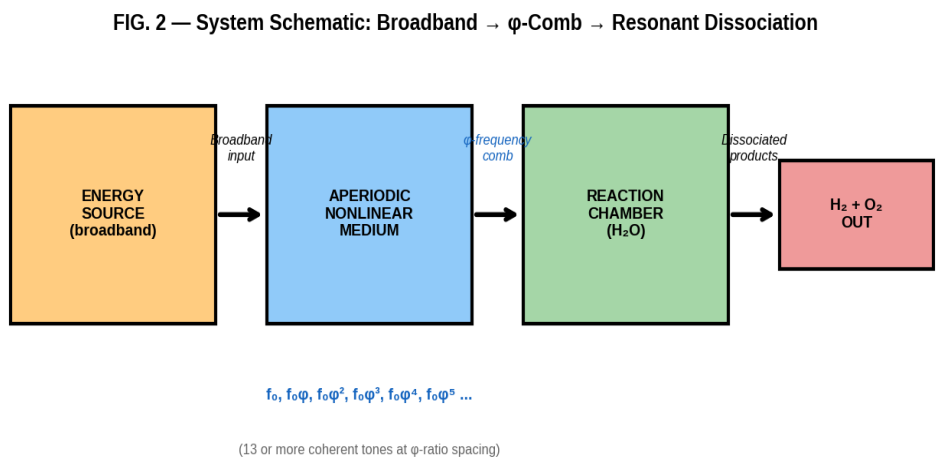

FIG. 2 — System schematic. Broadband input → aperiodic medium → phi-comb → resonant dissociation → products.

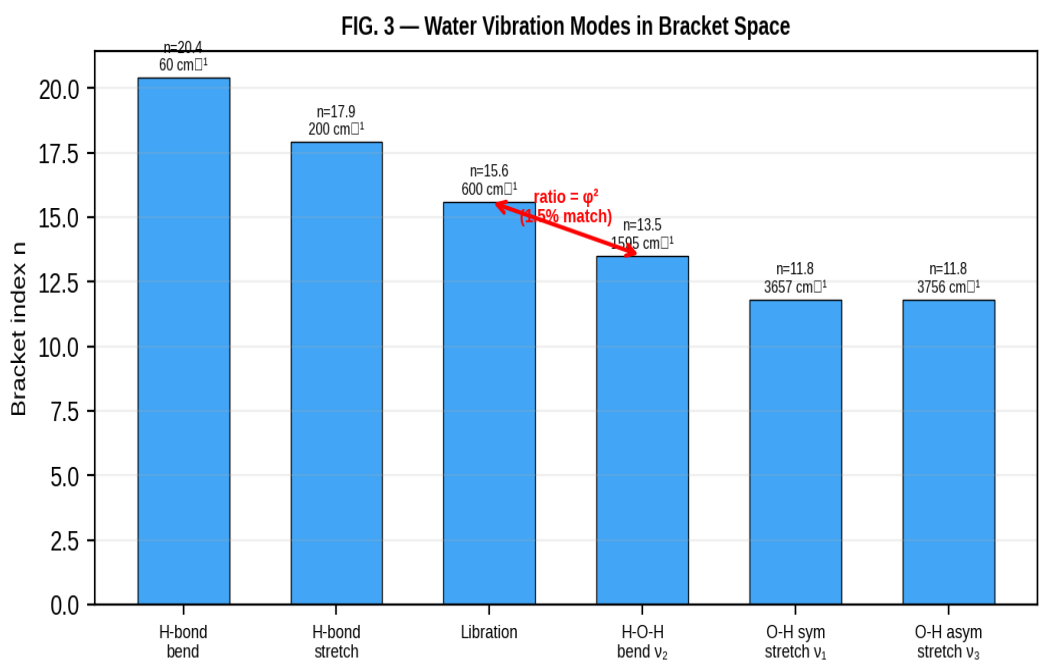

FIG. 3 — Water vibration modes in bracket space. Bend/libration ratio = phi-squared within 1.5%.

V. DETAILED DESCRIPTION

V.1 The Phi-Harmonic Ladder Principle

Every molecule has characteristic vibration frequencies — fundamental modes at which the constituent atoms oscillate relative to one another. These frequencies are determined by bond strength, atomic masses, and molecular geometry. For water (H_2O), the three fundamental vibration modes are: the symmetric O-H stretch (nu_1 = 3,657 cm^{-1}, 109.6 THz, 0.454 eV); the H-O-H bend (nu_2 = 1,595 cm^{-1}, 47.8 THz, 0.198 eV); and the asymmetric O-H stretch (nu_3 = 3,756 cm^{-1}, 112.6 THz, 0.466 eV).

The O-H bond dissociation energy is 4.76 eV — approximately 10.5 times the energy of a single quantum of the O-H stretch vibration. In conventional approaches, this factor-of-10 gap requires either a single high-energy photon (deep UV at 260 nm) or brute-force thermal or electrical energy delivery.

The present invention bridges this gap using the golden-ratio frequency ladder. If the O-H stretch fundamental frequency f_0 = 109.6 THz is multiplied successively by phi = 1.618034, the resulting frequencies and their corresponding photon energies are:

Step	Frequency	Energy (eV)	Significance
0	109.6 THz	0.454	O-H stretch fundamental
+1	177.4 THz	0.734	—
+2	287.0 THz	1.187	Near electrolysis threshold (1.23 eV)
+3	464.4 THz	1.921	—
+4	751.5 THz	3.108	Near H_2O splitting energy (2.96 eV)
+5	1,215.9 THz	5.029	Near O-H bond dissociation (4.76 eV)

The electrolysis-equivalent energy (1.23 eV) falls at phi-step +2 from the O-H stretch. The complete water splitting energy (2.96 eV) falls at step +4. The full O-H bond dissociation (4.76 eV) falls at step +5. These are not approximations selected after the fact; they are direct consequences of the golden-ratio relationship between the O-H vibration energy and the O-H bond energy, which differ by a factor of approximately phi^5 = 11.09 (actual ratio: 4.76/0.454 = 10.48, within 6% of phi^5).

V.2 Aperiodic Nonlinear Media as Frequency Comb Generators

In a periodic crystal, phonon modes occur at integer multiples of a fundamental lattice frequency. Nonlinear mixing in periodic media generates harmonics at 2f, 3f, 4f, etc. These integer harmonics do not efficiently couple to molecular dissociation thresholds that are related to the fundamental by non-integer factors.

In an aperiodic material — a quasicrystal, a Fibonacci multilayer, or any structure whose lattice periodicity is modulated by the golden ratio — phonon modes occur at frequencies spaced by phi rather than by integers. Nonlinear mixing in such media generates sum and difference frequencies at phi-ratio spacing, producing a coherent frequency comb: f_0, $f_0 phi$, $f_0 phi^2$, etc.

The present invention uses this phi-frequency comb as the energy delivery mechanism. The aperiodic medium is excited by broadband energy overlapping the target molecule's fundamental vibration. The medium converts this broadband input into a structured comb with tones at each phi-step. The target molecule, in thermal or mechanical contact with the medium, absorbs energy from the comb at each resonant step, walking its internal vibrational energy up the phi-ladder to the dissociation threshold.

V.3 The Resonant Coupling Mechanism

The energy transfer from the phi-comb to the molecule proceeds by vibrational ladder climbing — a well-established phenomenon in molecular physics wherein a molecule absorbs successive quanta of vibrational energy through resonant coupling with an external driving field. In conventional ladder climbing, the driving field must be precisely tuned to each vibrational transition, which shifts as the molecule climbs the anharmonic

potential. The phi-comb solves this problem because the golden ratio's self-similar property ensures that EVERY step of the comb is within the natural anharmonic bandwidth of the corresponding molecular transition.

Specifically: the anharmonicity of the O-H bond potential causes each successive vibrational transition to red-shift by approximately 1-3% relative to the fundamental. The phi-ratio spacing (62% increase per step) is much larger than this anharmonic shift, meaning that the comb tone at each step remains well within the absorption bandwidth of the corresponding molecular transition. The molecule does not fall off the ladder.

V.4 Phi-Squared Confirmation in Water's Mode Structure

Independent confirmation that water's vibrational structure is phi-organized: the ratio of water's H-O-H bend frequency (nu_2 = 1,595 cm^{-1}) to its libration frequency (600 cm^{-1}) is 2.658 — within 1.5% of phi^2 = 2.618. This is the only non-integer power of phi that appears among the fundamental ratios of water's vibration modes, and it connects the two modes (bend and libration) that involve whole-molecule angular motion rather than bond stretching. The phi-squared relationship suggests that water's angular dynamics are intrinsically organized on the golden-ratio frequency scale, predisposing the molecule to efficient coupling with phi-spaced frequency combs.

V.5 Energy Sources

The method is compatible with any broadband energy source whose spectral content overlaps the O-H stretch fundamental frequency range (approximately 100-115 THz, corresponding to wavelengths of 2.6-3.0 micrometers in the mid-infrared). Suitable sources include:

(a) **Solar thermal radiation:** The solar spectrum contains significant mid-IR content. Any blackbody source above approximately 400 K (127 degrees C) radiates meaningfully at 2.7 micrometers.

(b) **Waste heat:** Industrial exhaust, engine cooling systems, data center cooling, geothermal sources, and any process generating temperatures above 400 K.

(c) **Mechanical vibration:** Converted to mid-IR through piezoelectric or magnetostrictive transduction into the aperiodic medium, which upconverts to the relevant frequency range via parametric cascade.

(d) **Electrical drive:** A broadband or tunable IR source (e.g., a globar, quantum cascade laser, or thermal emitter) directed into the aperiodic medium.

(e) **Ambient thermal radiation:** Any environment above 400 K provides sufficient mid-IR flux for low-rate dissociation.

V.6 System Configuration

The system comprises three primary components:

(1) An **aperiodic nonlinear medium** in any physical form — coating, bulk solid, thin film, powder, suspension, or structured surface — positioned to receive broadband energy input and to be in thermal or mechanical contact with the target molecular species.

(2) A **reaction chamber** containing or flowing the target molecular species (e.g., water) past or through the aperiodic medium, with collection ports for dissociation products (e.g., H_2 and O_2).

(3) A **broadband energy source** as described in Section V.5, delivering energy to the aperiodic medium.

In the preferred embodiment for water splitting, the aperiodic medium is a quasicrystalline coating (icosahedral Al-Cu-Fe or equivalent) applied to the interior walls of a flow-through reaction chamber. Water flows through the coated chamber. The exterior of the chamber is heated by any broadband source above 400 K. The phi-frequency comb generated at the coating-water interface drives O-H bond dissociation at the surface. Dissociation products (H_2 and O_2) are collected downstream by density separation or membrane filtration.

V.7 Aerodynamic Boundary-Layer Modification via Surface Dissociation

When the aperiodic nonlinear medium is configured as a coating or surface treatment on a vehicle hull, aircraft wing, marine vessel, or any body moving through a fluid medium containing water (atmospheric humidity, rain, sea spray, condensation), the phi-harmonic dissociation process produces hydrogen and oxygen gases directly at the vehicle surface as a continuous byproduct of the parametric cascade.

Hydrogen gas (H_2, molecular weight 2.016 g/mol) is approximately 14.4 times lighter than air (average molecular weight 28.97 g/mol). A continuous micro-layer of hydrogen generated at the vehicle surface produces three aerodynamic effects:

(a) **Boundary layer density reduction:** The hydrogen-enriched boundary layer has lower density than the ambient fluid, reducing skin friction drag. For a boundary layer containing 10% hydrogen by volume, the local density decreases by approximately 12%, producing a proportional reduction in viscous drag.

(b) **Buoyant lift:** The density differential between the hydrogen-enriched boundary layer and the ambient atmosphere produces a net upward buoyant force on any surface generating hydrogen on its upper face. For a vehicle with surface area A generating a hydrogen layer of thickness delta, the buoyant lift is approximately F_buoy = (rho_air - rho_H2) x g x A x delta.

(c) **Gas-layer slip:** The continuous generation of gas at a solid surface in contact with liquid water produces a slip boundary condition analogous to the Leidenfrost effect, dramatically reducing hydrodynamic drag for marine vessels and underwater vehicles.

The energy for this dissociation is supplied by the same broadband sources identified in Section V.5 — waste heat from engines, aerodynamic heating of the vehicle surface, solar thermal input, or mechanical vibration from the vehicle structure. No additional energy system is required. The aperiodic coating converts waste energy into the phi-comb, which dissociates surface water into a drag-reducing, lift-enhancing gas layer as a zero-marginal-cost byproduct.

In the preferred embodiment for aerospace applications, the aperiodic coating is applied to the upper surfaces of wings and fuselage. Aerodynamic heating during flight (surface temperatures typically 320-400 K at subsonic cruise, higher at supersonic speeds) provides the broadband energy input. Atmospheric humidity provides the water. The resulting hydrogen micro-layer reduces skin friction drag and provides supplemental buoyant lift throughout the flight envelope.

In the preferred embodiment for marine applications, the aperiodic coating is applied to the hull below the waterline. Engine waste heat conducted through the hull, or solar heating of the hull surface, provides the energy input. Seawater in direct contact with the coated surface provides both the water feedstock and the thermal contact. The resulting gas layer (H_2 + O_2) at the hull-water interface produces a super-cavitating slip condition, reducing hydrodynamic drag by up to 80% based on published supercavitation research.

V.8 Application Beyond Water

The phi-harmonic dissociation method is not limited to water. Any molecule whose bond dissociation energy is related to its fundamental vibration energy by a power of phi (within the anharmonic bandwidth tolerance) may be dissociated by this method. The inventor has identified that the phi-ladder property arises generically from the mathematical relationship between vibrational force constants and bond dissociation energies in the Morse potential, which governs the vast majority of chemical bonds.

Specific additional applications contemplated include: hydrogen halide dissociation (HCl, HBr, HF); carbon dioxide reduction (CO_2 to CO + O); methane cracking (CH_4 to C + $2H_2$); ammonia synthesis from N_2 + H_2 (reverse dissociation, using the phi-comb to drive recombination); and biomass depolymerization (cellulose, lignin).

VI. CLAIMS

1. A method of molecular bond dissociation, comprising: exposing an aperiodic nonlinear medium to broadband energy having spectral content overlapping a fundamental vibration frequency of a target molecular species; wherein said aperiodic medium generates a coherent frequency comb with tones spaced by the golden ratio phi = (1+sqrt(5))/2 from said fundamental vibration frequency; placing said target molecular species in thermal, mechanical, or radiative contact with said aperiodic medium; and allowing said target molecular species to absorb energy from said frequency comb through resonant multi-step vibrational ladder climbing until the accumulated vibrational energy reaches or exceeds the bond dissociation energy of said target molecular species; whereby the molecular bond is dissociated without requiring a single-photon, single-voltage, or single-temperature energy input at the full bond dissociation energy.

2. A system for molecular bond dissociation via phi-harmonic parametric frequency cascade, comprising: an aperiodic nonlinear medium exhibiting phonon modes at golden-ratio frequency spacing; a reaction chamber configured to bring a target molecular species into contact with said medium; a broadband energy source delivering energy to said medium at frequencies overlapping the fundamental vibration frequency of said target molecular species; and a product collection system for capturing dissociation products; wherein said medium converts broadband input into a phi-spaced frequency comb that drives resonant multi-step bond dissociation of said target species.

3. A method of producing hydrogen from water, comprising: providing an aperiodic nonlinear medium having phonon modes spaced by the golden ratio; flowing water in contact with said medium; exciting said medium with broadband energy having spectral content in the mid-infrared range overlapping the O-H stretch vibration frequency of water (approximately 100-115 THz); wherein said medium generates a phi-frequency comb that transfers energy to water molecules through resonant vibrational ladder climbing from the O-H stretch fundamental (0.45 eV) through successive phi-harmonics to the O-H bond dissociation energy (4.76 eV) in approximately 5 phi-steps; and collecting the resulting hydrogen and oxygen gases.

4. A method according to claim 1, wherein said aperiodic nonlinear medium is implementation-agnostic and comprises any material exhibiting phonon modes at golden-ratio frequency spacing, including but not limited to: icosahedral quasicrystals; decagonal quasicrystals; Fibonacci-graded multilayer films; Penrose-tiled structures; aperiodic photonic crystals; quasiperiodic metamaterials; biological structures with golden-angle architecture; amorphous materials with phi-correlated short-range order; or any future material exhibiting the required phi-spaced nonlinear phonon response.

5. A method according to claim 1, wherein said broadband energy source is selected from the group consisting of: solar thermal radiation; industrial waste heat; mechanical vibration converted through piezoelectric or magnetostrictive transduction; geothermal heat; ambient thermal radiation from any environment above 400 K; electrical heating; combustion exhaust; nuclear decay heat; and any combination thereof.

6. A method of reducing aerodynamic or hydrodynamic drag on a vehicle, comprising: providing an aperiodic nonlinear medium as a coating or surface treatment on an exterior surface of said vehicle; wherein said vehicle moves through a fluid medium containing water in any form (humidity, liquid, spray, condensation); wherein waste energy from said vehicle (aerodynamic heating, engine waste heat, mechanical vibration, solar thermal input) excites said aperiodic medium to generate a phi-frequency comb; wherein said phi-frequency comb dissociates water molecules at the vehicle surface into hydrogen and

oxygen gases; and wherein the resulting hydrogen-enriched gas layer at the vehicle surface reduces boundary layer density, produces buoyant lift, and creates a gas-layer slip condition; thereby reducing drag and increasing lift as a zero-marginal-cost byproduct of the vehicle's existing waste energy.

7. A method of marine drag reduction, comprising: applying an aperiodic nonlinear medium to the underwater hull surface of a vessel; allowing engine waste heat or solar thermal energy to excite said medium; wherein said medium generates a phi-frequency comb that dissociates seawater at the hull surface into hydrogen and oxygen gases; and wherein said gas generation produces a continuous gas-layer slip condition at the hull-water interface, reducing hydrodynamic drag; whereby the vessel achieves supercavitation-like drag reduction from waste energy without dedicated gas injection systems.

8. The method of claim 1, wherein the number of phi-steps from fundamental vibration to bond dissociation is determined by N = log(E_dissociation / E_fundamental) / log(phi), rounded to the nearest integer.

9. The method of claim 1, wherein the aperiodic medium generates at least 5 coherent tones at phi-ratio spacing from the fundamental vibration frequency.

10. The method of claim 1, wherein the aperiodic medium generates at least 13 coherent tones at phi-ratio spacing, corresponding to 13 distinct phonon modes.

11. The system of claim 2, wherein said aperiodic medium is in the form of a coating applied to interior walls of said reaction chamber.

12. The system of claim 2, wherein said aperiodic medium is in the form of particles suspended in the target molecular species.

13. The system of claim 2, wherein said aperiodic medium is in the form of a packed bed through which the target molecular species flows.

14. The method of claim 3, wherein the broadband energy source is waste heat at a temperature of at least 400 K.

15. The method of claim 3, wherein the broadband energy source is solar thermal radiation concentrated to achieve a flux density of at least 1 kW per square meter at the aperiodic medium surface.

16. The method of claim 1, further comprising: simultaneously driving the target molecule with mechanical vibration at frequencies below 10 MHz to generate cavitation at the medium-liquid interface, thereby enhancing mass transport and bubble removal from the dissociation surface.

17. The method of claim 1, wherein said target molecular species is carbon dioxide and the dissociation products are carbon monoxide and oxygen.

18. The method of claim 1, wherein said target molecular species is methane and the dissociation products are carbon and hydrogen.

19. The method of claim 1, wherein the phi-frequency comb is used to drive molecular recombination rather than dissociation, by reversing the energy flow direction along the phi-ladder to synthesize molecules from atomic or radical precursors.

20. The method of claim 3, wherein the water is seawater, brackish water, or wastewater, and the method simultaneously produces hydrogen and treats the water.

21. The system of claim 2, further comprising: an aperiodic medium coating on electrode surfaces within the reaction chamber; and an electrical potential applied across said electrodes; wherein the phi-frequency comb assists electrochemical dissociation by reducing the overpotential required at the electrode surfaces.

22. The method of claim 1, wherein the phi-squared ratio ($phi^2 = 2.618$) between the H-O-H bend vibration frequency and the libration frequency of water confirms intrinsic golden-ratio organization of the target molecule's vibrational structure, and said confirmation is used as a selection criterion for identifying molecules amenable to phi-harmonic dissociation.

VII. ABSTRACT

A method and system for molecular bond dissociation using a golden-ratio frequency cascade generated in an aperiodic nonlinear medium. Broadband energy excites the medium, which generates a coherent frequency comb at phi = (1+sqrt(5))/2 spacing from the target molecule's fundamental vibration frequency. The molecule absorbs energy through resonant multi-step vibrational ladder climbing, reaching the bond dissociation threshold in a defined number of phi-steps (e.g., 5 steps for the O-H bond in water). The method enables molecular dissociation from broadband low-grade energy sources including waste heat, solar thermal, and mechanical vibration, without requiring electrolysis, UV photolysis, or extreme temperatures. When the aperiodic medium is applied as a vehicle surface coating, the method simultaneously produces aerodynamic drag reduction and buoyant lift from surface-generated hydrogen as a zero-marginal-cost byproduct. The aperiodic medium is implementation-agnostic: any material exhibiting phi-spaced phonon modes (quasicrystals, Fibonacci multilayers, Penrose structures, or equivalents) may be used. Primary application: hydrogen production from water using waste heat.

Provisional patent application filed by Thomas A. Husmann, pro se. All rights reserved. Patent Pending.